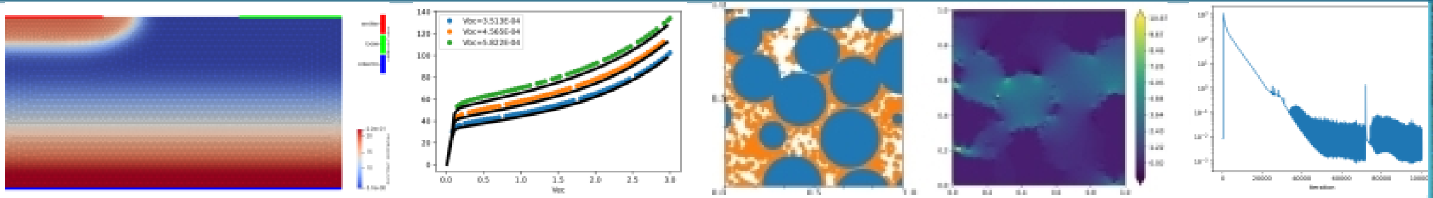
SAND2024-09558C

Development and testing of subcircuit surrogate models using Data-Driven Exterior Calculus and Xyce-PyMi



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* at the time of this work

WCCM-PANACM 2024
Minisymposium 1816: Data-driven device and circuits models
July 26th, 2024

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Surrogate modeling motivation and approach

Jump right in – importance of surrogate modeling addressed by other talks.
Example hierarchy of system modeling and simulation (top to bottom/low-fi to high-fi)

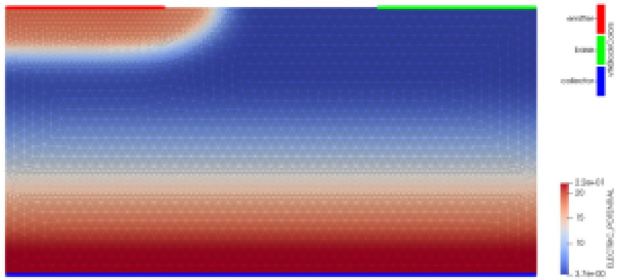
- System (electrical, mechanical, thermal)
- Electromagnetic
- Circuit
- Device

Observations:

- Desirable to develop technique which can interface between multiple levels.
- Commonality: respect physics conservation principles! (flux, energy, current, …)
- Many approaches fix a domain and subsequently learn a function on that domain; Instead, try to learn **both the domain and the function** while respecting conservation regardless of the domain parameterization.
- Inspiration from device compact modeling, which takes subject matter expertise to map local physics to parameterized lumped or ideal device components into a compact model circuit topology.

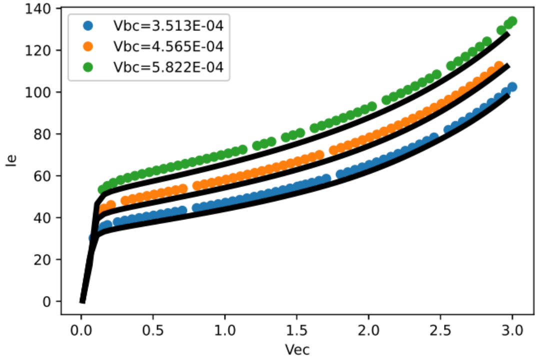
Physics-Informed, Rapid and Automated MachIne-learning for compact model Development (PIRAMID) approach

Goal: machine-learned circuit simulator compatible semiconductor device model from TCAD & data



Fine-scale, high-fidelity data

Goal



Xyce-PyMi callable DDEC model

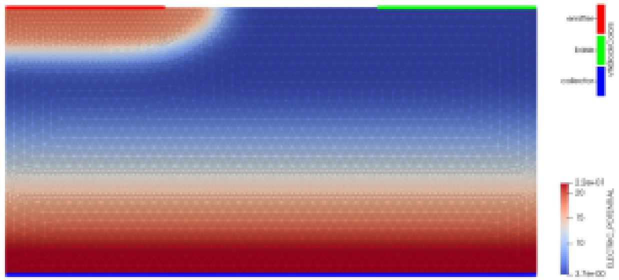
Trask, Nathaniel & Huang, Andy & Hu, Xiaozhe. (2022)
Enforcing exact physics in scientific machine learning: A data-driven exterior calculus on graphs
Journal of Computational Physics. 456. 110969. 10.1016/j.jcp.2022.110969.

Huang, Andy & Trask, Nathaniel & Gao, X. & Reza, Shahed & Patel, Ravi & Brissette, Christopher & Hu, Xiaozhe. (2022)
Continuum semiconductor physics model compression via Data-driven Discrete Exterior Calculus 10.2172/2003428.
8th European Congress on Computational Methods in Applied Sciences and Engineering held June 5-9, 2022 in Oslo, Norway

Huang, A., Trask, N., Brissette, C., & Hu, X. (2021).
Association for the Advancement of Artificial Intelligence - Machine Learning for the Physical Sciences (AAAI-MLPS) 2021
Greedy Fiedler Spectral Partitioning for Data-driven Discrete Exterior Calculus. *March 22-24, 2021 in Palo Alto, CA, United States.*.

Physics-Informed, Rapid and Automated MachIne-learning for compact model Development (PIRAMID) approach

Goal: machine-learned circuit simulator compatible semiconductor device model from TCAD & data
Tool stack: Charon (TCAD), Data-driven Discrete Exterior Calculus (Python/Tensorflow), Xyce-PyMi
Generation: Charon (TCAD) → physics & data-driven coarsening: local 1-D I-V data → DDEC
Usage: pickled DDEC model → Xyce-PyMi



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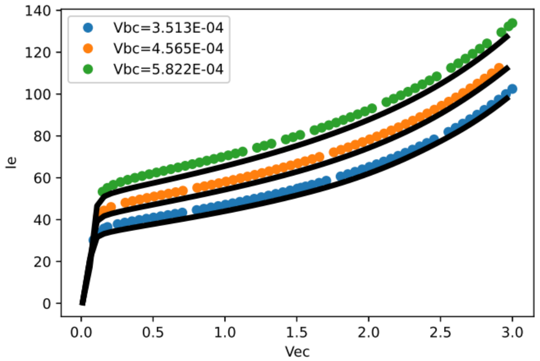

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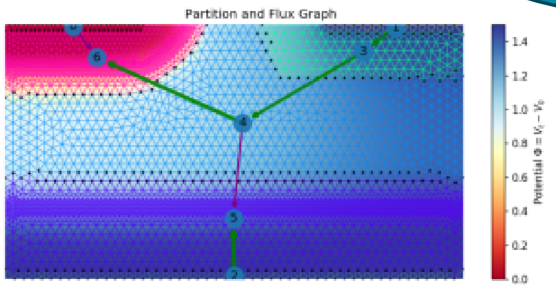
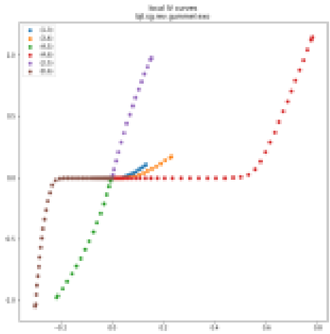


Coarse-grained data



```
DDEC 2D BJT terminal contact DC sweep I-V characteristics
V2 0 2 1
V1 0 1 5.814e-4
YGENEXT ddec 0 1 2 dummymodel
+ SPARAMS={NAME=MODULENAME, DATAFILE VALUE=../xyce_model_BJT.py,
    ./data/xyce_BJT.pick}
.model dummymodel genext
.DC V2 0.01 3.0 0.05
.print DC V(2) I1(YGENEXT!ddec) I2(YGENEXT!ddec) I3(YGENEXT!ddec)
.END
```

Machine-learned DDEC model

Conservation principles in a machine learning framework

Conservation laws relating current and voltages are matrix-vector product identities on the edge and nodal quantities of the graph of the circuit.

Current i = edge vector (oriented)
Voltage v = edge vector (oriented)
Potential ϕ = nodal scalar

A = incidence matrix
(branches to nodes)

Kirchhoff Voltage Law: $v \in im(A^T)$, $v = A^T \phi$
Kirchhoff Current Law: $i \in ker(A)$, $Ai = 0$
Tellegen's theorem: $(v, i) = 0$

Seek a graph domain (on which to learn edge-wise I(V) functions which reproduces data) and machine learning framework which respects conservation laws during training and prediction.

Data-driven Discrete Exterior Calculus (DDEC)

On graph of nodes and edges,
p, q denote potential, current
f boundary data.

$$X = \begin{bmatrix} X_{\text{nodes}} \\ X_{\text{edges}} \end{bmatrix} = \begin{bmatrix} p \\ q \end{bmatrix}$$

$$f = \begin{bmatrix} f_{\text{nodes}} \\ f_{\text{edges}} \end{bmatrix}$$

Goal:
learn q(p) edge I(V) model while respecting KCL and boundary conditions

$$\begin{aligned} \mathcal{NN}_e\left(\text{grad } p; \boldsymbol{\xi}\right)|_e &= q|_e \\ \text{div } q|_n &= 0 \text{ for all } n \in \Omega_{\text{interior}} \\ p|_n &= p_{\text{bc}}(n) \text{ for all } n \in \Omega_{\text{bdy}} \end{aligned}$$

$$\begin{aligned} \text{div } q|_n &= 0 \text{ for all } n \in \Omega_{\text{interior}} \\ p|_n &= p_{\text{bc}}(n) \text{ for all } n \in \Omega_{\text{bdy}} \\ \mathcal{NN}\left(\text{grad } p; \boldsymbol{\xi}\right) - q &= 0 \end{aligned}$$

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Adler, James & Cavanaugh, Casey & Hu, Xiaozhe & Huang, Andy & Trask, Nathaniel. (2023)
A Stable Mimetic Finite-Difference Method for Convection-Dominated Diffusion Equations
SIAM Journal on Scientific Computing. 45. A2973-A3000. 10.1137/22M1514337.

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$$\text{div } q|_n = 0 \text{ for all } n \in \Omega_{\text{interior}}$$
$$p|_n = p_{\text{bc}}(n) \text{ for all } n \in \Omega_{\text{bdy}}$$

Transforms potentially multi-dimensional V→I map of multiple contact device to a family of 1D problems satisfying global constraint of a conservation law.
Physics-based q(p) form can be used, perturbed by NN.

Unlike PINNs which enforce physics by penalty,
DDEC allows direct formulation of conservation law

Specialized DDEC
to circuit models

$$\text{div } q|_n = 0 \text{ for all } n \in \Omega_{\text{interior}}$$
$$p|_n = p_{\text{bc}}(n) \text{ for all } n \in \Omega_{\text{bdy}}$$
$$\mathcal{NN}\left(\text{grad } p; \boldsymbol{\xi}\right) - q = 0$$

$$\Downarrow$$

$$\begin{bmatrix} \mathcal{L}_{\text{int nodes}}[X] \\ \mathcal{L}_{\text{bdy nodes}}[X] \\ \mathcal{L}_{\text{edge info}}[X] \end{bmatrix} = \begin{bmatrix} 0 \\ f_{\text{bdy nodes}} \\ 0 \end{bmatrix}$$

$$\Downarrow$$

$$L(X, \boldsymbol{\xi}, \boldsymbol{\lambda}) = ||X - X_{\text{data}}||^2_{\ell^2} + \boldsymbol{\lambda}^\top(\mathcal{L}[X, \boldsymbol{\xi}] - f)$$

$$\left.\begin{aligned} \nabla_X L = 0 \\ \nabla_{\boldsymbol{\lambda}} L = 0 \\ \nabla_{\boldsymbol{\xi}} L = 0 \end{aligned}\right\} \to \boldsymbol{\xi}$$

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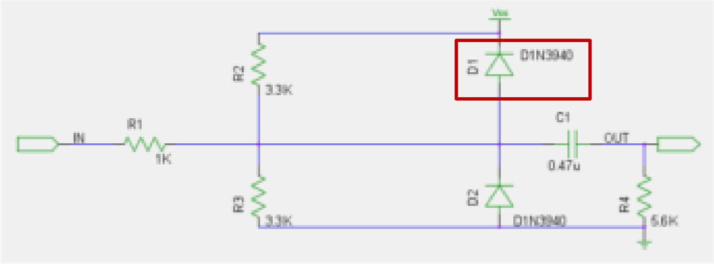
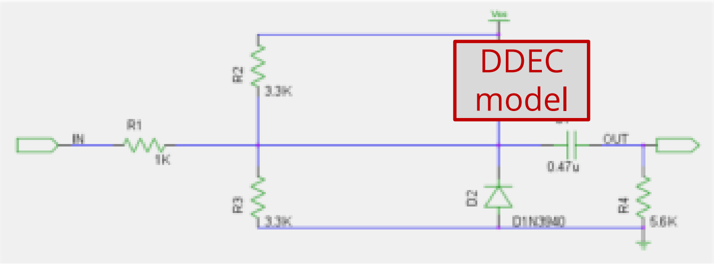
DDEC applied to device modeling for use in circuit

Hypothetical situation: in a circuit design iteration, how does selection of a new diode affect circuit performance? If TCAD data exists on that diode, train and use a DDEC model.

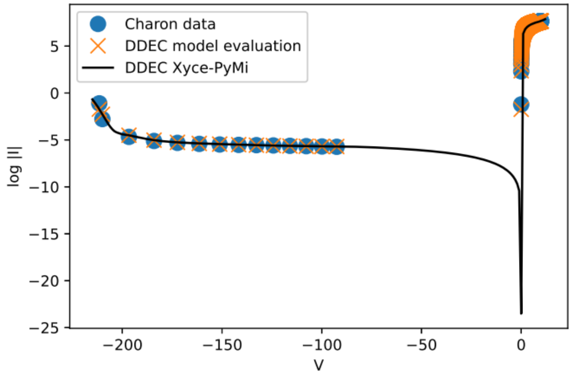



```
Diode Clipper Circuit with DC sweep
* Voltage Sources
VCC 1 0 5V
VIN 3 0 0V
* Diodes
D1 2 1 D1N3940
D2 0 2 D1N3940
* Resistors
R1 2 3 1K
R2 1 2 3.3K
R3 2 0 3.3K
R4 4 0 5.6K
* Capacitor
C1 2 4 0.47u
.MODEL D1N3940 D
.DC VIN -15 15 1
.PRINT DC V(1) V(2) V(3) V(4)
.END
```

Replace D1 with DDEC model of a different diode, trained on TCAD data




```
DDEC model in Diode Clipper Circuit with DC sweep
* Voltage Sources
VCC 1 0 5V
VIN 3 0 0V
* Diodes
YGENEXT ddec 1 2
+ SPARAMS={NAME=MODULENAME, DATAFILE VALUE=../xyce_model.py,
   ./data/xyce_GaNdiode_py37_2.pick}
D2 0 2 D1N3940
* Resistors
R1 2 3 1K
R2 1 2 3.3K
R3 2 0 3.3K
R4 4 0 5.6K
* Capacitor
C1 2 4 0.47u
.MODEL D1N3940 D
.DC VIN -10 5 1
.PRINT DC V(1) V(2) V(3) V(4)
.END
```



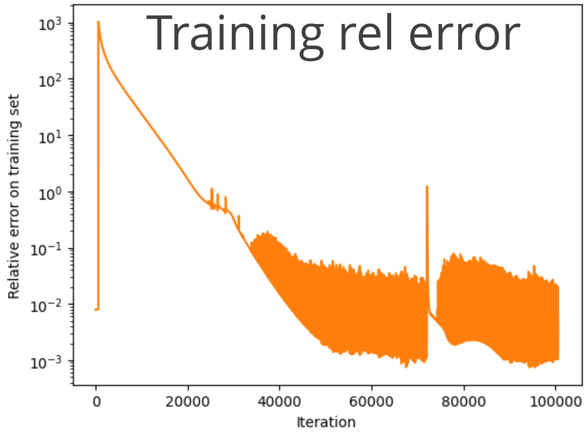
DDEC applied to subcircuit modeling

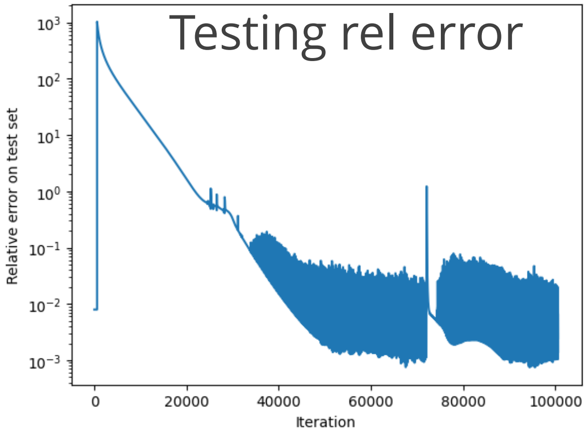
- Treat subcircuit as a “device”
- No physics coarsening – fix a graph and train on the terminal I(V) data
- DC sweep analysis performed on the large circuit to obtain subcircuit port I(V) data

Ongoing work: extend transient DDEC framework for use in subcircuit modeling

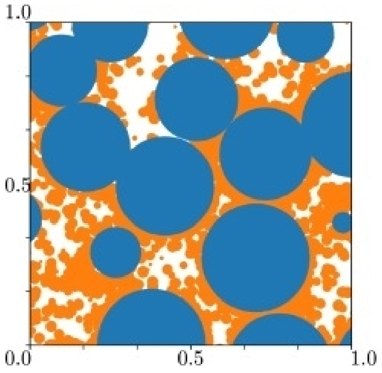
```
********* Diode Clipper Circuit ******************
V1 1 0 1V
R1 2 0  1K
X1 1 2 CLP
.subckt CLP IN OUT
D1 IN   OUT  D1N3940
C1 OUT   N1    10p
R1 N1    0     1K
.ends
.MODEL D1N3940 D(  IS = 4E-10
+                  RS = .105
+                   N = 1.48
+                  TT = 8E-7
+                 CJO = 1.95E-11
+                  VJ = .4
+                   M = .38
+                  EG = 1.36
+                 XTI = -8
+                  KF = 0
+                  AF = 1
+                  FC = .9
+                  BV = 600
+                 IBV = 1E-4)

***************************************************
.DC LIN V1 0.00 1.00 0.025
.PRINT DC FORMAT=CSV FILE=CLPSWEEP.prn V(*) I(*)
.END
```

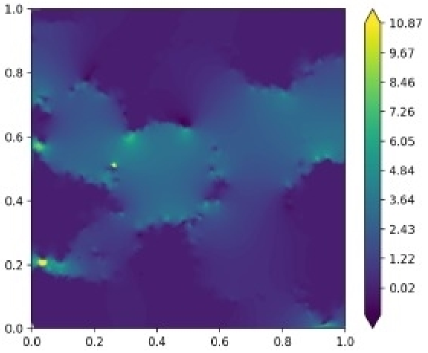
```
********* DDEC Diode Clipper Circuit ******************************

V1 1 0 1V
R1 2 0  1K
YGENEXT ddec_clipper_subckt 1 2 ddec_model
.SPARAMS={NAME=MODULENAME, DATAFILE VALUE=./ddec.py,./subckt.pick}

.model ddec_model genext

*******************************************************************
.DC LIN V1 0.00 1.00 0.025
.PRINT DC FORMAT=CSV FILE=CLPSWEEP_DDEC.prn V(*) I(*)

.END
```

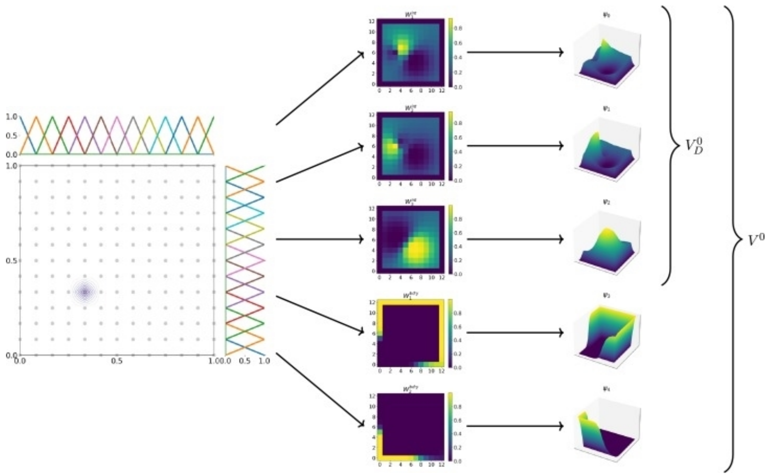


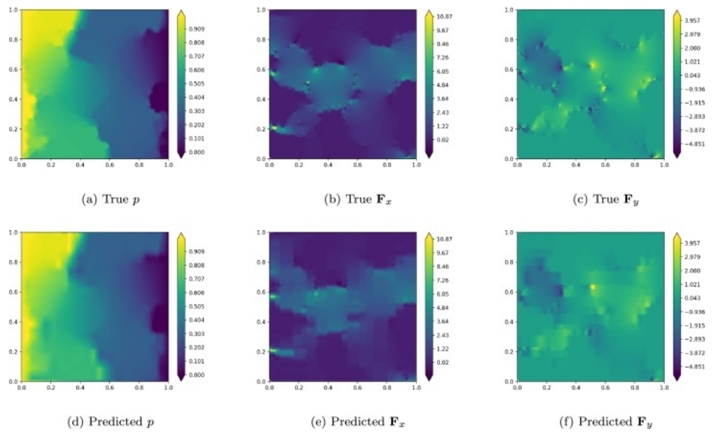


Future considerations: Whitney form DDEC

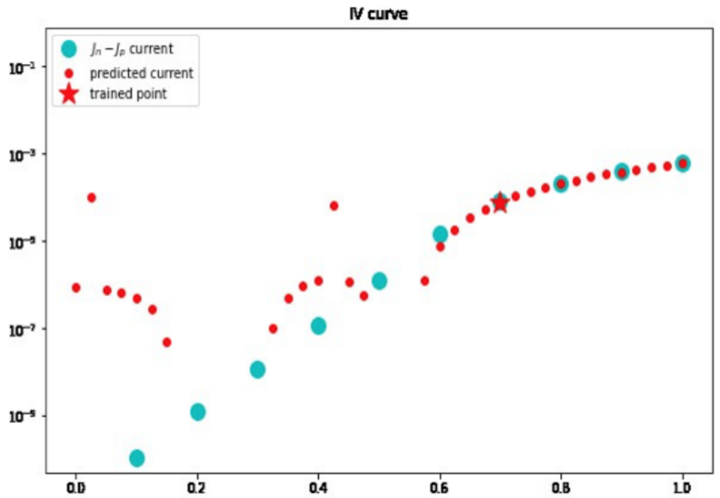








- Avoids coarsening, start with complete graph connectivity → every graph is a subgraph of K_n
- Learn the spatial domain and basis functions most suitable to the divergence-form conservation law
- Transient capability being developed



Actor, Jonas & Hu, Xiaozhe & Huang, Andy & Roberts, Scott & Trask, Nathaniel. (2023).
Data-Driven Whitney Forms for Structure-Preserving Control Volume Analysis
Journal of Computational Physics. 496. 112520. 10.1016/j.jcp.2023.112520.